LLPS-Score: Predicting Protein Phase Separation and Biological Function from Physicochemical Features

BTP report submitted in partial fulfillment of the requirements for the Degree of B.Tech. in Computer Science & Applied Mathematics
and Computer Science & Artificial Intelligence

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Student's Declaration

We hereby declare that the work presented in the report entitled **"LLPS-Score: Predicting Protein Phase Separation and Biological Function from Physicochemical Features"** submitted by us for the partial fulfillment of the requirements for the degree of Bachelor of Technology in Computer Science & Applied Mathematics and Computer Science & Artificial Intelligence at Indraprastha Institute of Information Technology, Delhi, is an authentic record of our work carried out under guidance of Dr. Garima Rani & Dr. Ganesh Bhimanna Bagler.

Due acknowledgements have been given in the report to all material used. This work has not been submitted anywhere else for the reward of any other degree.

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This is to certify that the above statement made by the candidates is correct to the best of our knowledge.

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Abstract

Liquid-liquid phase separation (LLPS) has emerged as a key principle driving the organization of biochemical networks in cells, generating dynamic, membrane-less condensates that compartmentalize biochemical reactions. Precise prediction of LLPS-prone proteins is crucial for understanding cellular compartmentalization and elucidating disease-associated mechanisms. In this report, we develop a high-throughput computational pipeline to predict LLPS propensity using only primary amino acid sequences. We curated a balanced and diverse dataset of 6,289 protein sequences, extracting explicit physicochemical features such as intrinsic disorder (IDR), low-complexity regions (LCR), hydrophobicity, and Shannon entropy. To capture covert positional patterns, we applied representation learning, converting sequences into dense numerical vectors using Word2Vec with overlapping 3-grams. Leveraging these combined features, we designed and trained a Convolutional Neural Network (Hybrid CNN). Our deep learning architecture achieves exceptional predictive performance, demonstrating a test accuracy of 96.74% and an ROC-AUC of 0.982. This study highlights that combining protein linguistics with convolutional networks provides a highly scalable, robust, and interpretable tool for the computational screening of phase-separating proteins.

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Chapter 1

Introduction

Liquid-liquid phase separation (LLPS) has emerged as an important mechanism by which cells organize biomolecules into dynamic, membrane-less condensates. These condensates play a central role in regulating processes such as transcription, RNA metabolism, stress response, and intracellular signaling. Because LLPS is influenced by subtle sequence-level and physicochemical properties, identifying phase-separating proteins experimentally is both time-consuming and difficult to scale across large proteomes. This creates a strong need for computational methods that can predict LLPS propensity directly from amino acid sequence information.

In this project, we aim to capture the hidden molecular grammar that governs phase separation by treating protein sequences as biological text. To support this, we expanded our data curation pipeline and constructed a balanced dataset of 6,289 protein sequences, enabling more robust learning and better generalization across different protein classes. Rather than relying only on manually designed rules, we combine explicit biochemical features with learned sequence representations so that the model can leverage both known biophysical signals and latent contextual patterns.

For representation learning, the protein sequences were tokenized into overlapping 3-grams and used to train a Word2Vec model. This allowed us to generate dense embedding matrices that preserve local contextual relationships between amino acids, much like word embeddings in natural language processing. These embeddings were then concatenated with explicit global features such as sequence length, GRAVY score, and LCR/IDR fractions to form a comprehensive numerical representation of each protein. This hybrid encoding captures both global composition and local sequential context, making it especially suitable for downstream deep learning.

We then feed these combined representations into a Convolutional Neural Network (CNN). The convolutional layers are particularly well suited for this task because they can slide across the embedded sequence space and automatically learn useful patterns without manual feature selection. In this way, the model can detect motif-like structures, residue dependencies, and short-range interaction patterns that may contribute to LLPS behavior. With this approach, the model achieved an ROC-AUC of 0.982, showing that sequence information alone contains strong signals relevant to phase separation.

1.1 Motivation and Deep Learning Approach

The motivation behind this work is to develop a predictive framework that goes beyond conventional feature-based classification and learns directly from the inherent structure of protein sequences. LLPS is not determined by a single residue type or simple compositional threshold; rather, it emerges from a combination of sequence length, disorder, charge distribution, low-complexity regions, and context-dependent residue arrangements. Because of this complexity, deep learning offers a natural advantage by being able to model non-linear relationships and hierarchical patterns in the input data.

To robustly capture these signatures, we constructed a balanced dataset of 6,289 protein sequences and represented each sequence using a combination of learned embeddings and engineered biochemical descriptors. The Word2Vec-based 3-gram embeddings allow the model to view proteins in a language-like form, where local sequence fragments carry contextual meaning. At the same time, global features such as GRAVY, LCR fraction, and IDR fraction ensure that known physicochemical indicators of phase separation are explicitly retained in the input representation. This combination gives the model a richer and more informative description of the protein than either representation alone.

The CNN was chosen because it can detect local patterns within the embedded sequence space in a way that is analogous to motif recognition. As the convolutional filters move across the input, they learn important subsequence patterns and feature combinations that may correspond to LLPS-related biological signals. This makes the model capable of identifying subtle sequence organization principles that are difficult to capture using traditional classifiers. In addition, the architecture is scalable, efficient, and well suited for large datasets, making it practical for proteome-wide prediction tasks.

The strong performance of this approach, including an ROC-AUC of 0.982, demonstrates that raw sequence information alone can encode meaningful biophysical properties associated with phase separation. More importantly, it supports the broader idea that LLPS-related behavior is not arbitrary, but is embedded in the sequence grammar of proteins. This provides both a useful computational tool and a biologically interpretable framework for understanding phase separation from first principles.

Chapter 2

Literature Review

2.1 Biological Foundations of Liquid-Liquid Phase Separation (LLPS)

Liquid-liquid phase separation (LLPS) has emerged as a fundamental principle governing the spatial organization of cellular biochemistry. Unlike traditional membrane-bound organelles, cells also form membrane-less compartments, such as nucleoli, stress granules, and P-bodies, through phase separation. These structures enable the cell to compartmentalize reactions dynamically, allowing rapid assembly and disassembly in response to environmental cues.

At the molecular level, LLPS is driven by multivalent interactions among proteins and nucleic acids. These interactions are typically weak and transient—such as electrostatic interactions, π–π stacking, cation–π interactions, and hydrophobic effects—but collectively lead to the formation of dense liquid-like condensates. The process can be understood through polymer physics, where proteins behave as associative polymers capable of forming reversible interaction networks.

A defining characteristic of these condensates is their dynamic and liquid-like behavior, which allows continuous exchange of molecules with the surrounding environment. However, this same property also makes LLPS highly sensitive to perturbations. Dysregulation of phase separation has been implicated in several diseases, particularly neurodegenerative disorders such as ALS, Alzheimer's disease, and frontotemporal dementia, where liquid condensates can transition into solid, pathogenic aggregates.

Given this biological significance, accurately predicting which proteins can undergo LLPS is an important and challenging problem. Since experimental validation is resource-intensive, computational approaches that infer LLPS propensity directly from sequence data are highly valuable.

2.2 Sequence Determinants: IDRs, LCRs, and Physicochemical Properties

The ability of a protein to undergo phase separation is strongly encoded in its primary amino acid sequence, particularly through features that influence flexibility, interaction density, and compositional bias. Two key sequence characteristics associated with LLPS are Intrinsically Disordered Regions (IDRs) and Low-Complexity Regions (LCRs).

IDRs are segments that do not fold into stable three-dimensional structures. Instead, they remain flexible and adopt multiple conformations, enabling them to participate in transient, multivalent interactions. This structural flexibility allows proteins to behave like dynamic polymers, facilitating the formation of liquid-like condensates.

LCRs are regions with biased amino acid composition, often enriched in a small subset of residues such as glycine, serine, glutamine, and asparagine. These regions typically exhibit repetitive patterns and reduced sequence complexity. LCRs frequently overlap with IDRs and contribute to phase separation by promoting weak, distributed interactions across the sequence.

In addition to these structural features, several global physicochemical properties play a crucial role:

- **Hydrophobicity (GRAVY score):** LLPS-prone proteins tend to have lower hydrophobicity, indicating reduced tendency to form stable folded cores and increased solvent exposure.
- **Sequence Length:** Longer proteins provide more interaction sites, supporting multivalency and increasing the likelihood of phase separation.
- **Shannon Entropy:** Lower sequence complexity, reflected by reduced entropy, is often associated with repetitive and biased sequences typical of phase-separating proteins.

These properties form the foundation of feature engineering in computational models. By quantifying disorder, compositional bias, and global sequence characteristics, it becomes possible to capture biologically meaningful signals that contribute to LLPS propensity.

2.3 Traditional Computational Predictors for LLPS

Early computational approaches to LLPS prediction relied primarily on rule-based and statistical methods. These models focused on identifying specific sequence features associated with phase separation, such as prion-like domains, disordered regions, or enrichment of interaction-prone residues.

Many of these methods used sliding window techniques to analyze local sequence segments, estimating the likelihood of interactions such as π–π stacking or electrostatic attraction. While these approaches provided valuable insights into the biophysical basis of LLPS, they were inherently limited in scope.

A major limitation of traditional predictors is their reliance on explicit human-designed rules. These rules are often derived from known examples and may not generalize well to diverse protein families. Additionally, sliding window approaches primarily capture local patterns and fail to account for long-range dependencies and global sequence organization, which are critical for multivalent interactions.

As a result, these models often struggle with the heterogeneity of LLPS proteins and exhibit limited predictive performance when applied to large and diverse datasets. This highlights the need for more flexible, data-driven approaches capable of learning complex patterns directly from sequence data.

2.4 Representation Learning: Treating Proteins as Biological Text

A more recent and powerful approach to sequence analysis involves treating protein sequences as a form of biological language. In this framework, amino acids are analogous to words, and sequences correspond to sentences, allowing the application of Natural Language Processing (NLP) techniques.

One of the most widely used methods in this context is Word2Vec, which learns vector representations of tokens based on their contextual co-occurrence. By tokenizing protein sequences into overlapping k-mers (e.g., 3-grams), it becomes possible to learn embeddings that capture local sequence context and relationships between residues.

These embeddings encode contextual and positional information that goes beyond simple frequency-based features. For example, they can capture:

- Repeating motifs and sequence patterns
- Local residue interactions
- Context-dependent functional similarities

Unlike handcrafted features, which are limited to predefined metrics, embedding-based representations allow the model to learn latent sequence structure directly from data. This enables the detection of subtle and complex patterns that are difficult to capture using traditional approaches.

In practice, combining these embeddings with engineered biochemical features results in a hybrid representation, where explicit knowledge and implicit learning complement each other. This forms the basis for more powerful predictive models.

2.5 Deep Learning and Convolutional Neural Networks in Bioinformatics

Deep learning models have become increasingly important in bioinformatics due to their ability to learn hierarchical and non-linear representations from complex data. Among these models, Convolutional Neural Networks (CNNs) are particularly effective for analyzing sequential inputs.

While CNNs are commonly associated with image processing, 1D CNNs are well suited for biological sequences. In this setting, convolutional filters slide across the input sequence or embedding space, detecting local patterns and motifs. These filters act as learnable feature extractors, automatically identifying relevant subsequences without manual intervention.

The hierarchical structure of CNNs allows them to capture patterns at multiple scales. Early layers detect simple local features, while deeper layers combine these into more complex representations. This makes CNNs especially effective for modeling the distributed and context-dependent nature of LLPS signals.

By applying convolution over sequence embeddings, CNNs can identify:

- Motif-like residue patterns
- Interaction-prone regions

- Local dependencies and sequence organization

A key advantage of this approach is that it does not rely on predefined biological rules. Instead, the model learns directly from data, enabling it to generalize across diverse protein sequences.

In this work, a hybrid deep learning framework is employed, where CNN-based feature extraction is combined with explicit global physicochemical features. This integration allows the model to leverage both known biological principles and automatically learned sequence patterns, resulting in a robust and accurate predictor of LLPS propensity.

Chapter 3

Methodology

3.1 Dataset Construction and Merging

To develop a robust and generalizable predictive framework, a large-scale dataset comprising 6,289 protein sequences was curated by integrating multiple LLPS-related sources. The dataset includes both LLPS-positive sequences and a carefully constructed negative control set consisting of proteins not known to undergo phase separation. This balanced composition ensures that the model learns discriminative patterns rather than dataset-specific biases.

During preprocessing, all sequences were standardized to a consistent format. Entries containing non-standard amino acids were removed to maintain compatibility with downstream feature extraction and embedding pipelines. Duplicate sequences were eliminated to prevent redundancy and potential data leakage.

Handling missing values is critical in high-throughput biological datasets. Any undefined or missing feature values (NaNs), arising from feature computation or parsing inconsistencies, were imputed to zero, ensuring numerical stability without introducing artificial bias.

To facilitate efficient neural network training, all numerical features were standardized using a pre-trained scaler, ensuring zero mean and unit variance. Importantly, the scaler was not re-fitted on validation or test data, thereby preserving strict separation between training and evaluation sets and preventing data leakage. This preprocessing pipeline ensures that the dataset is clean, consistent, and suitable for both machine learning and deep learning models.

3.2 Feature Engineering

Phase separation is governed by underlying physicochemical principles that are reflected in sequence composition and structural flexibility. To capture these determinants, we extracted a set of explicit features that encode both global and local biochemical properties of protein sequences. These features serve as interpretable signals that complement the data-driven representations learned later.

3.2.1 Global Biochemical Features

For each protein sequence, three primary global descriptors were computed:

- **Sequence Length:** This metric captures the multivalency potential of the protein. Longer sequences are more likely to contain multiple interaction sites ("stickers"), increasing the probability of forming phase-separated condensates.
- **Hydrophobicity (GRAVY Score):** The Grand Average of Hydropathy (GRAVY) score was calculated to estimate the overall hydrophobic or hydrophilic nature of the protein. LLPS-prone proteins typically exhibit lower hydrophobicity, reflecting reduced structural compaction and increased solvent exposure.
- **Shannon Entropy:** Shannon entropy was computed over the amino acid distribution to quantify sequence complexity. Lower entropy values indicate repetitive or biased compositions, which are commonly associated with low-complexity regions involved in phase separation.

These features provide a compact yet informative representation of global sequence characteristics relevant to LLPS.

3.2.2 LCR and IDR Fraction Analysis

In addition to global properties, LLPS is strongly influenced by structural disorder and compositional bias. To capture these effects, we quantified:

- **Low-Complexity Region (LCR) Fraction:** The proportion of residues belonging to low-complexity regions, which are enriched in repetitive and interaction-prone motifs.
- **Intrinsically Disordered Region (IDR) Fraction:** The fraction of residues predicted to be disordered, reflecting the protein's conformational flexibility and ability to engage in transient multivalent interactions.

These features provide a direct measure of the protein's polymer-like behavior, which is a key requirement for phase separation. Together, LCR and IDR fractions serve as critical indicators of a protein's propensity to form dynamic condensates.

3.3 Representation Learning

While explicit features capture global and biologically interpretable properties, they are insufficient to model the local sequence context and positional dependencies that govern residue interactions. To address this limitation, we employ a representation learning strategy inspired by Natural Language Processing.

3.3.1 Overlapping 3-Gram Tokenization

Protein sequences were transformed into a sequence of overlapping 3-grams (k-mers of length 3). This tokenization approach preserves the local neighborhood context of each residue, allowing the model to capture short-range dependencies and motif-like structures.

Unlike single-residue encoding, overlapping k-mers provide richer contextual information, enabling the representation of interaction-relevant subsequences that contribute to multivalent binding and phase separation.

3.3.2 Word2Vec Sequence Embeddings

The tokenized sequences were used to train a Word2Vec model, which learns distributed vector representations of 3-grams in an unsupervised manner. Each token is mapped to a dense vector in a high-dimensional embedding space, where proximity reflects contextual similarity.

This embedding process captures:

- Co-occurrence patterns of amino acid triplets
- Local structural and functional motifs
- Context-dependent residue relationships

By aggregating these embeddings, each protein sequence is represented as a structured matrix that encodes its latent sequence grammar. These embeddings provide a powerful alternative to handcrafted features, enabling the model to learn complex and non-linear relationships directly from sequence data.

3.4 Model Architecture

To effectively leverage both engineered features and learned embeddings, we designed a multi-modal architecture that integrates traditional machine learning and deep learning paradigms.

3.4.1 Baseline Classifiers

As an initial step, the engineered features were evaluated using classical ensemble methods such as Random Forest and AdaBoost. These models are well suited for tabular data and provide strong baseline performance due to their ability to model non-linear relationships.

However, these approaches are limited in their ability to process high-dimensional sequential data, such as embedding matrices. They treat features independently and cannot capture spatial dependencies inherent in sequence representations, which restricts their effectiveness for this task.

3.4.2 Hybrid Convolutional Neural Network (CNN)

To overcome these limitations, we developed a Hybrid 1D Convolutional Neural Network (CNN) that processes both embedding-based and explicit features simultaneously.

The architecture consists of two parallel branches:

- **Embedding Branch:** The Word2Vec embedding matrices are passed through multiple 1D convolutional layers. These layers apply learnable filters that slide across the sequence, acting as motif detectors. The network automatically learns spatial patterns such as residue clusters and interaction-prone segments. Pooling layers reduce dimensionality, while dropout layers improve generalization.
- **Explicit Feature Branch:** The standardized global features (sequence length, hydrophobicity, entropy, IDR/LCR fractions) are passed directly into fully connected layers, allowing the model to utilize biologically interpretable descriptors without distortion.

The outputs from both branches are concatenated to form a unified representation. This combined feature vector is then processed through additional dense layers, followed by dropout-based regularization. The final layer uses a sigmoid activation function to output a probability score representing LLPS propensity.

This hybrid design enables the model to simultaneously capture:

- Global physicochemical properties
- Local sequence patterns
- Context-dependent interactions

3.5 Model Inference and Pipeline Execution

The complete pipeline is designed as a high-throughput prediction system that integrates feature extraction, representation learning, and model inference into a unified workflow.

Given a raw protein sequence:

1. The feature extraction module computes global descriptors such as sequence length, GRAVY score, Shannon entropy, and IDR/LCR fractions.
2. The NLP module converts the sequence into overlapping 3-grams and generates Word2Vec embeddings.
3. These parallel inputs are fed into the Hybrid CNN, which processes both representations simultaneously.
4. The model outputs a probability score (0–1) indicating the likelihood of phase separation, which is then thresholded to produce a binary classification.

This modular design allows efficient scaling to large datasets and enables proteome-wide screening applications.

To further understand the compositional characteristics of the dataset, we analyzed the global distribution of amino acids across all sequences. This analysis provides insight into underlying biases in residue composition, which are closely linked to phase separation behavior and support the feature engineering strategy.

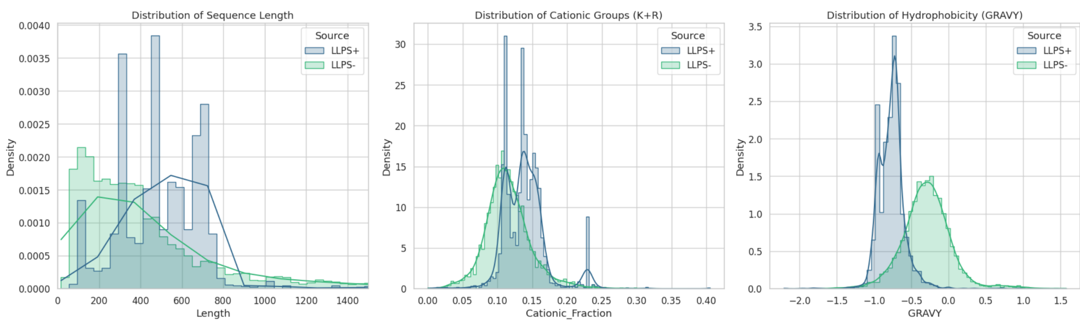

Figure 3.1: Frequency distribution of amino acids across the merged dataset (n=6,289).

Chapter 4

Results

4.1 Biochemical Feature Distributions

While traditional predictive models for protein phase separation often rely on explicit feature engineering—such as calculating net charge, hydrophobicity, and intrinsically disordered region (IDR) propensity—this dataset exhibits the complex, multi-dimensional signatures typical of liquid-liquid phase separation (LLPS) proteins. The baseline distributions of these biochemical properties highlight the inherent difficulty in using strict thresholds for classification, as LLPS+ and Non-LLPS proteins often share overlapping physicochemical characteristics in their sequence data. This overlap necessitates a more nuanced, high-dimensional approach to pattern recognition.

4.2 Implicit Feature Learning via Sequence Embeddings

To bypass the limitations of manual biochemical feature extraction, the predictive pipeline leverages Word2Vec embeddings. By treating amino acid sequences as linguistic constructs, the model successfully captures the contextual, latent biochemical properties of the proteins. These sequence embeddings map the complex biological motifs into a dense vector space, allowing the downstream network to implicitly learn the positional dependencies and structural grammar that drive phase separation, rather than relying on disparate, pre-calculated sequence metrics.

4.3 Predictive Performance and Evaluation

The effectiveness of the learned embeddings was evaluated by passing them through the classification models. The performance metrics highlight the models' ability to generalize and accurately distinguish phase-separating proteins from the negative dataset.

4.3.1 CNN Training Dynamics

The Convolutional Neural Network (CNN) demonstrated robust learning capabilities when processing the sequence embeddings.

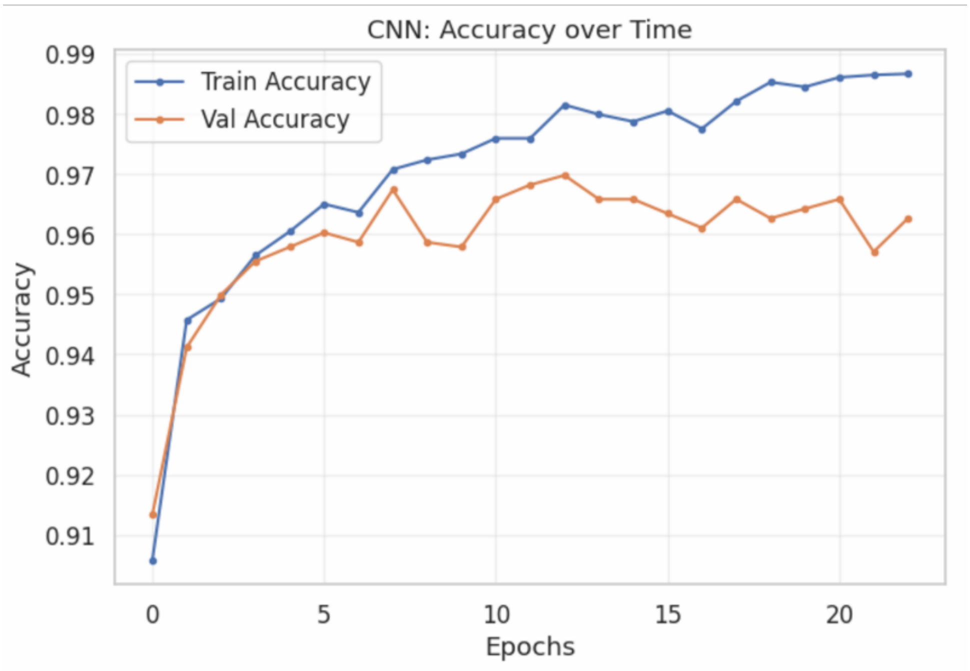

Figure 4.1: CNN training and validation accuracy across 22 epochs.

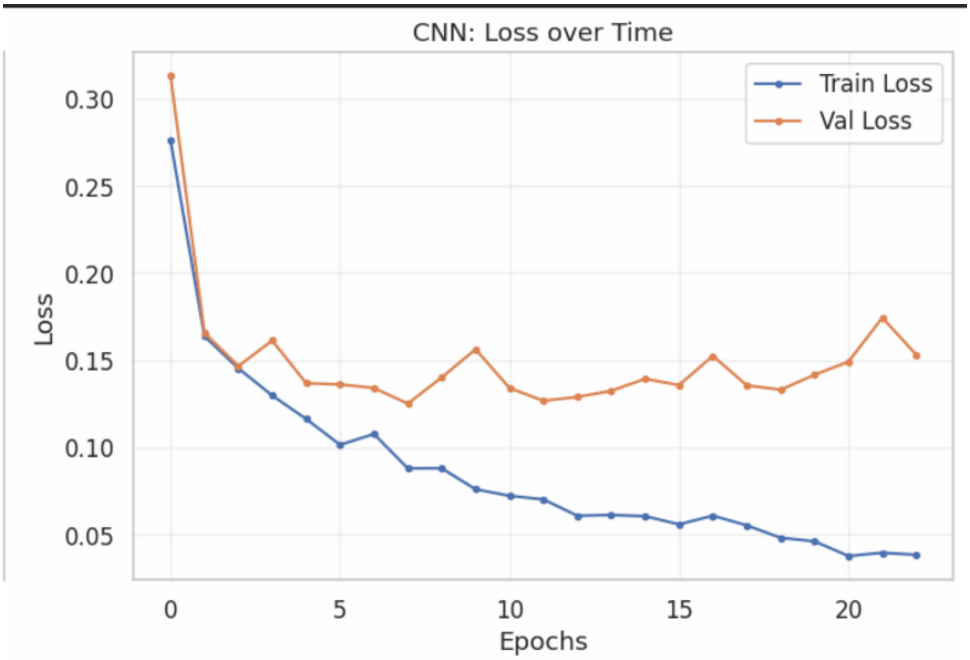

Figure 4.2: CNN training and validation loss over time.

As observed in the training dynamics (Figures 4.1 and 4.2), the CNN converges rapidly within the first 5 to 7 epochs. The training accuracy climbs steadily to approximately 0.985, while the validation accuracy stabilizes effectively in the 0.96 to 0.965 range. Looking at the loss curves, the validation loss reaches its minimum around epoch 11 (dropping to near 0.125). While the training loss continues to decrease, the validation loss shows a slight, gradual increase after epoch 15. This mild divergence indicates the onset of minor overfitting in the later epochs, suggesting that early stopping around epoch 12–15 would yield the most optimally generalized CNN model.

4.3.2 Receiver Operating Characteristic (ROC) Analysis

To assess the discriminative thresholding of the CNN, a Receiver Operating Characteristic (ROC) analysis was conducted.

The CNN achieves an excellent Area Under the Curve (AUC) of 0.982 (Figure 4.3). The steep incline of the True Positive Rate against a minimal False Positive Rate indicates that the network, powered by the spatial feature extraction of the convolutional layers over the embeddings, is highly confident in its predictions and maintains a strict separation boundary between the LLPS+ and Non-LLPS classes.

4.3.3 Baseline vs. CNN Performance Comparison

To contextualize the CNN's performance, it was compared against a baseline classification architecture.

The baseline model's training dynamics over 30 epochs (Figures 4.4 and 4.5) show a slightly more erratic validation curve compared to the CNN. The validation accuracy similarly hovers around 0.96, but the validation loss exhibits more volatility, fluctuating between 0.12 and 0.16 in the latter half of the training cycle.

Despite the volatility in training loss, the overall aggregated evaluation of the baseline model proved exceptionally strong (Figure 4.6). The baseline achieved an AUC of 0.99, slightly outperforming the CNN's 0.982. Furthermore, the confusion matrix reveals highly accurate class separation: 798 True Negatives against only 4 False Positives, alongside 415 True Positives and 41 False Negatives.

While the CNN offers a more stable loss trajectory and powerful spatial feature extraction, the baseline model's marginally superior AUC and strong specificity suggest that the Word2Vec embeddings themselves carry highly separable predictive power, regardless of the complexity of the downstream classifier. Future iterations could focus on hyperparameter tuning (such as increasing dropout or adjusting learning rates) to push the CNN beyond the baseline's benchmark.

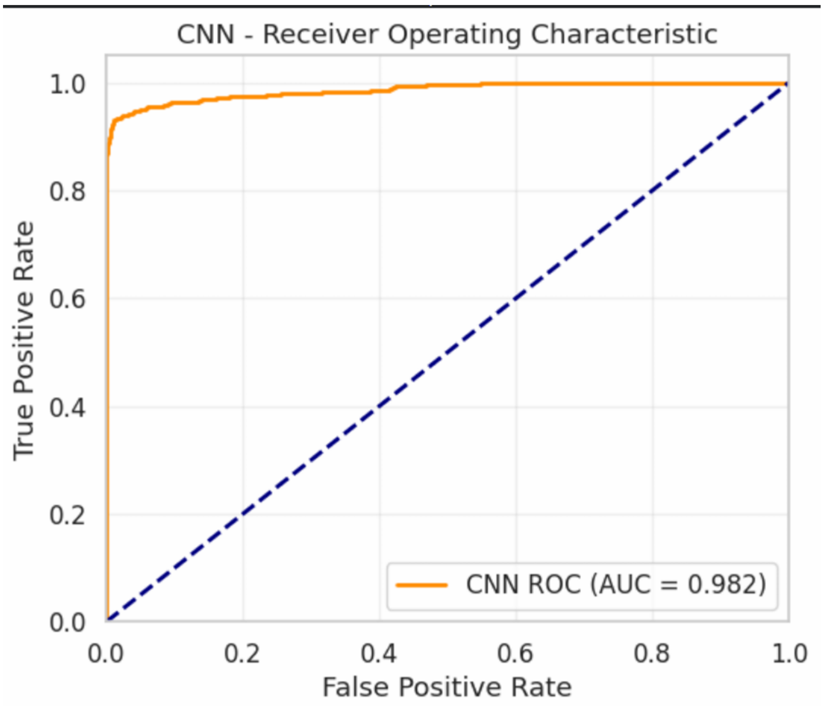

Figure 4.3: ROC curve for the CNN model.

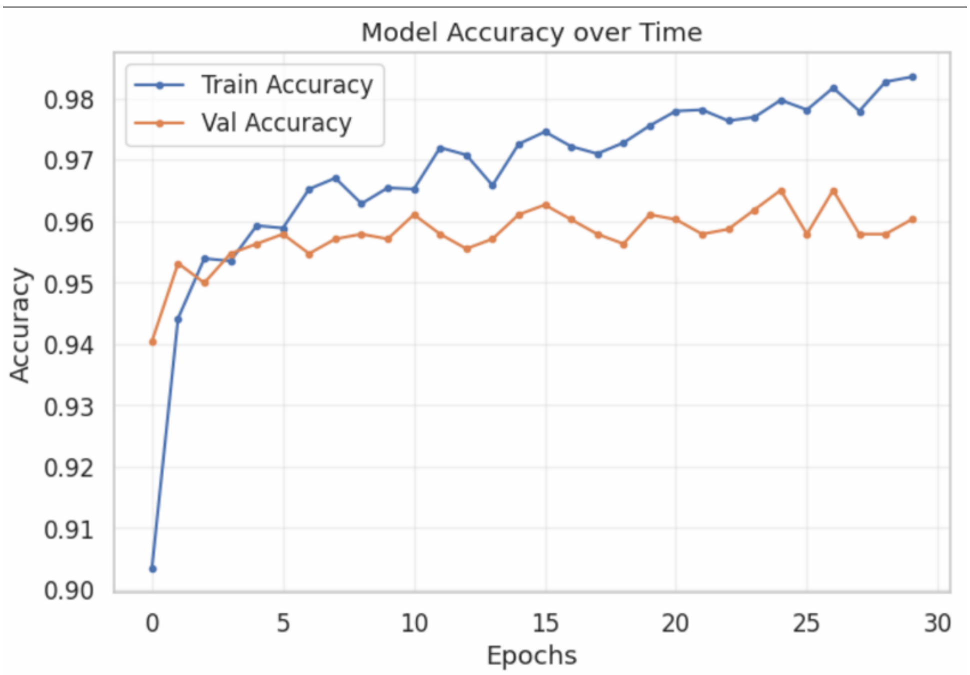

Figure 4.4: Baseline model training and validation accuracy across 30 epochs.

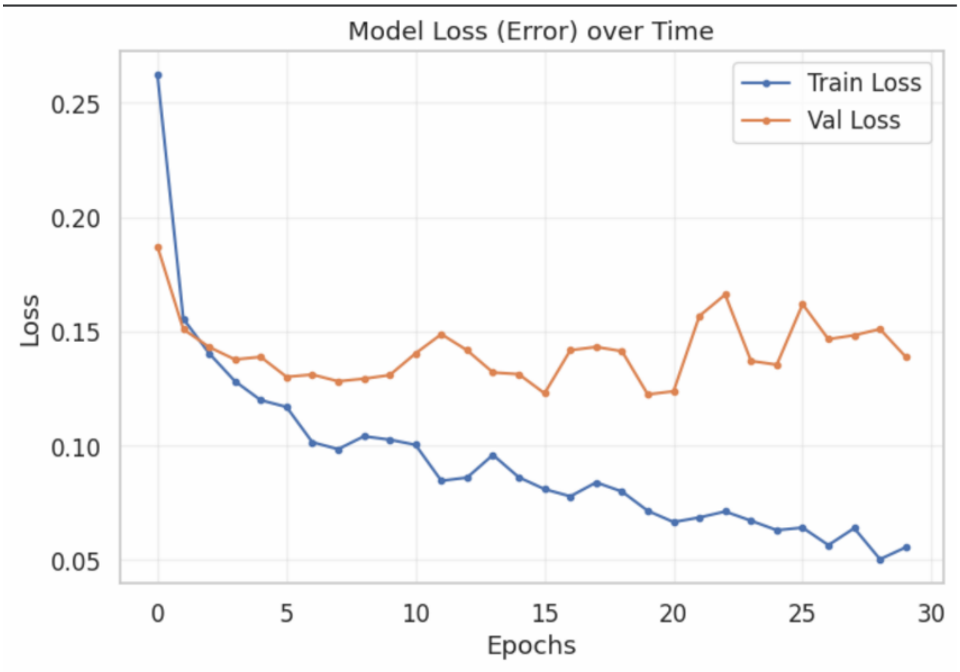

Figure 4.5: Baseline model training and validation loss.

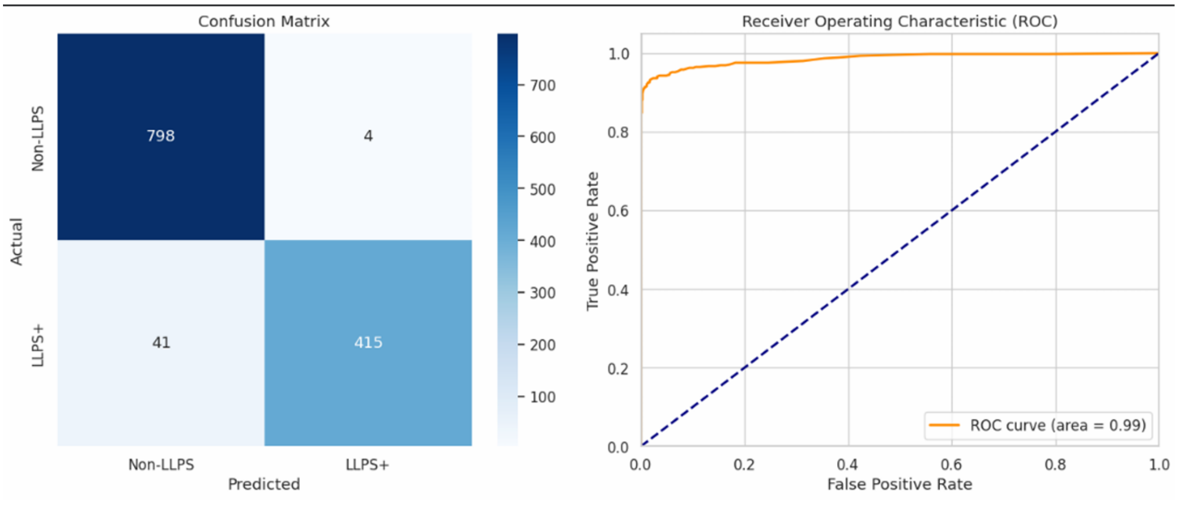

Figure 4.6: Confusion Matrix and ROC curve for the baseline model.

Chapter 5

Future Work

5.1 Future Goals

The present work establishes a strong computational foundation for LLPS prediction, but it can be extended into a practical and user-friendly platform with broader utility for researchers. In the next phase, we aim to convert the current model into a market-ready prediction product that can be used interactively by biologists, bioinformaticians, and experimental researchers.

5.1.1 Development of a User Interface

A major future goal is to build an intuitive graphical user interface (GUI) or web-based application for the model. This interface will allow users to interact with the system without needing to run code manually.

- Users will be able to enter a single protein sequence directly into the interface.
- Users will also be able to upload a CSV file containing multiple protein sequences for batch prediction.
- The system will process the input automatically and return classification results in a readable format.

5.1.2 LLPS Prediction and Probability Scoring

The interface will not only classify a protein as LLPS or non-LLPS, but will also provide a probability score indicating the confidence of phase separation.

- Each sequence will be assigned a probability of undergoing LLPS.
- The output will help users interpret borderline cases more carefully instead of relying only on binary labels.
- This probabilistic output can be especially useful for prioritizing candidates for experimental validation.

5.1.3 Feature-Level Output and Interpretation

Another important goal is to make the system more interpretable by reporting the key sequence-level properties used in prediction.

- The tool will display hydrophobicity (GRAVY score) for each sequence.
- It will also compute Shannon entropy to measure sequence complexity.
- Additional extracted features such as sequence length, IDR fraction, LCR fraction, and other relevant physicochemical properties will also be shown.
- This will allow users to understand not just the prediction, but also the biological reasons behind it.

5.1.4 Batch Screening for Large-Scale Analysis

To make the model more useful for real-world research, the application will support high-throughput screening of protein datasets.

- Researchers will be able to upload large CSV files containing many protein sequences.
- The tool will process them efficiently and generate a results file with predictions and feature values.
- This will be useful for proteome-wide screening and candidate discovery.

5.1.5 Productization and Deployment

A final future direction is to transform the project into a deployable scientific product.

- The model can be integrated into a streamlined web platform.
- Prediction results can be exported as downloadable reports or CSV files.
- The system can be extended with visualization modules such as probability plots, feature summaries, and sequence statistics.
- This would make the project accessible as a practical research tool rather than only a backend model.

Overall, these future developments will turn the current LLPS prediction pipeline into a complete end-to-end application that supports sequence input, batch analysis, probability-based prediction, and feature interpretation in a single platform.

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